Association for Project Management

Version #	Date	Details of change	Page / slide no.

Requirements management

Learning objective:

16. Understand requirements management as the ability to capture and monitor the requirements of a project.

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16a) Understand how to establish scope through requirements management processes (such as gather, analysis, justifying requirements and baseline needs).

Requirements management processes

Requirements management is an ongoing process throughout the project life cycle and should be commenced in the initial phase of the project. It facilitates the development of an agreed scope management. It is important to go through these processes:

Gather

- After having identified key stakeholders, information needs to be gathered from these to ensure that all possible requirements are gathered, recorded and documented.
- Where some stakeholders are missed from these early discussions, this can lead to important information being missed or to conflict further down the line.

Analysis

- Function analysis or function cost analysis will provide detail on the value that the requirements will contribute to the organisation.
- Analysis should enable any gaps to be identified or conflicting requirements resolved. This can also provide information on alternative solutions.
- The outcomes should be fed back to stakeholders and consensus reached

Justify requirements

- Functional requirements should be prioritised, linking directly back to the benefits, using techniques such as MoSCoW (M – must have; S – should have; C – could have; W – won't have).
- Reporting on the outcomes to stakeholders is then required to mitigate any misunderstandings or dispute later.

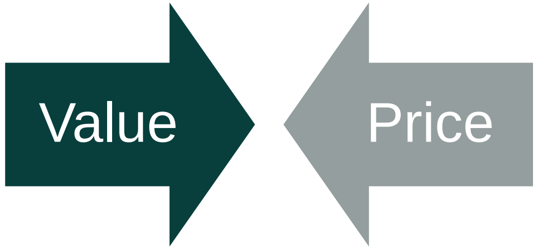
Baseline needs

- Once the functional requirements have been set, these form the baseline requirements for the project.
- From these, the project manager and project team can determine the project deliverables, manage change and mitigate scope creep.

Function analysis or function cost analysis

- We should only accept benefits at a cost that makes them worthwhile to the organisation, thus creating value. Value must be created at the beginning of the project through the development of the business case and in the selection of the option to be delivered, and then be maintained throughout the project.

- Function analysis and function cost analysis are value-based techniques which allow the analysis of the requirements and their business drivers by combining information from functions such as schedule management and investment appraisal. Analysing the time and cost required to deliver the requirements will result in thorough understanding of requirements and the value they will contribute to the objectives.



- The results of the analysis are communicated through individual consultation or group workshops. The product owner has a vital role in ensuring that requirements are always relevant to business needs and advises on the acceptance criteria necessary to achieve this.

- Value represents the amount of benefit that will accrue from the requirement and can be used to justify its priority – how important this requirement is compared to others.

Prioritisation techniques

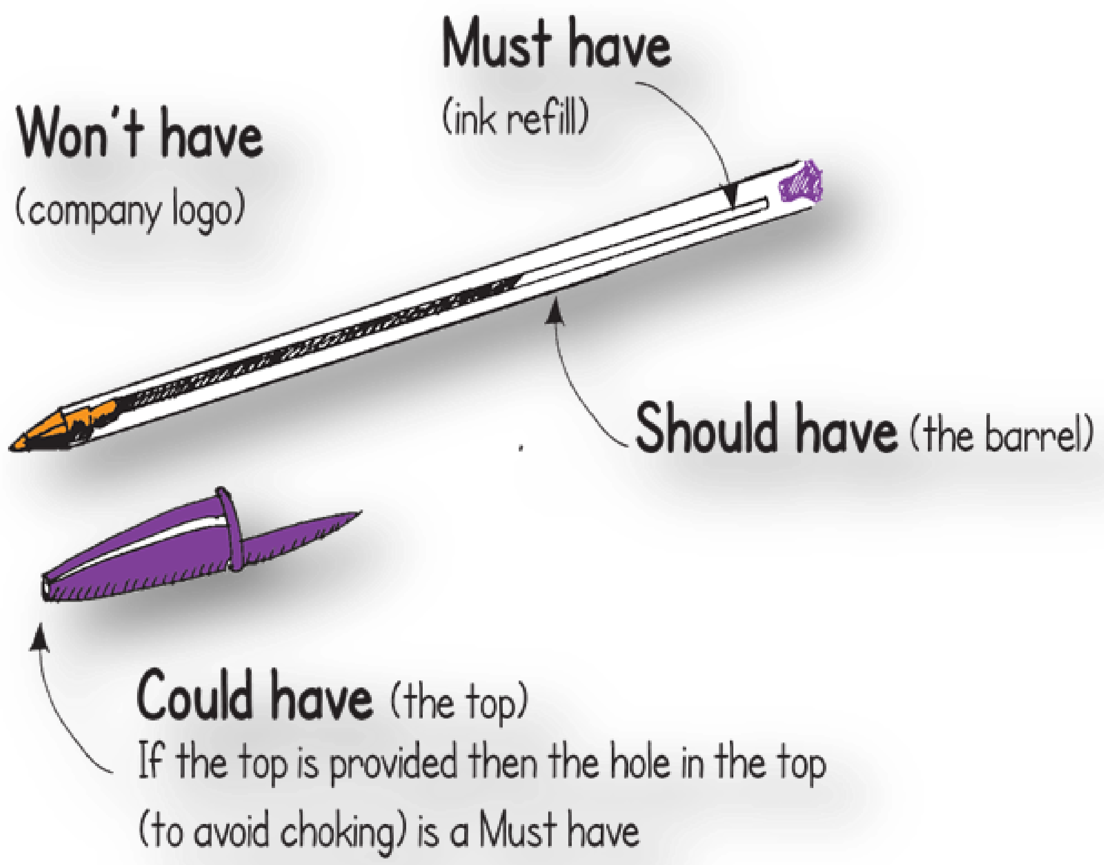
- Sometimes, more requirements are requested than it is feasible to deliver, so a prioritisation exercise is needed to highlight the most essential requirements and justify why the chosen requirements should be the ones that are baselined. A common prioritisation technique is the MoSCoW approach:
    - (M) Must have.
    - (S) Should have.
    - (C) Could have.
    - (W) Won't have.
- If using an agile methodology, the 'could have' and 'should have' requirements could be sacrificed if at any time the project was predicted to go over budget or be late. The 'must haves' would only be reduced as a very last resort.
- The main benefit of the process is to ensure that essential requirements are understood as an input to the work to select the optimal solution and then define the detailed scope of work to be delivered with acceptance criteria.
- This level of rigour mitigates the risk that, later in the life cycle, there will be dispute between the project and the organisation about completion and transition of the deliverables into use.

Photo credit: Brett Jordan

Prioritisation techniques

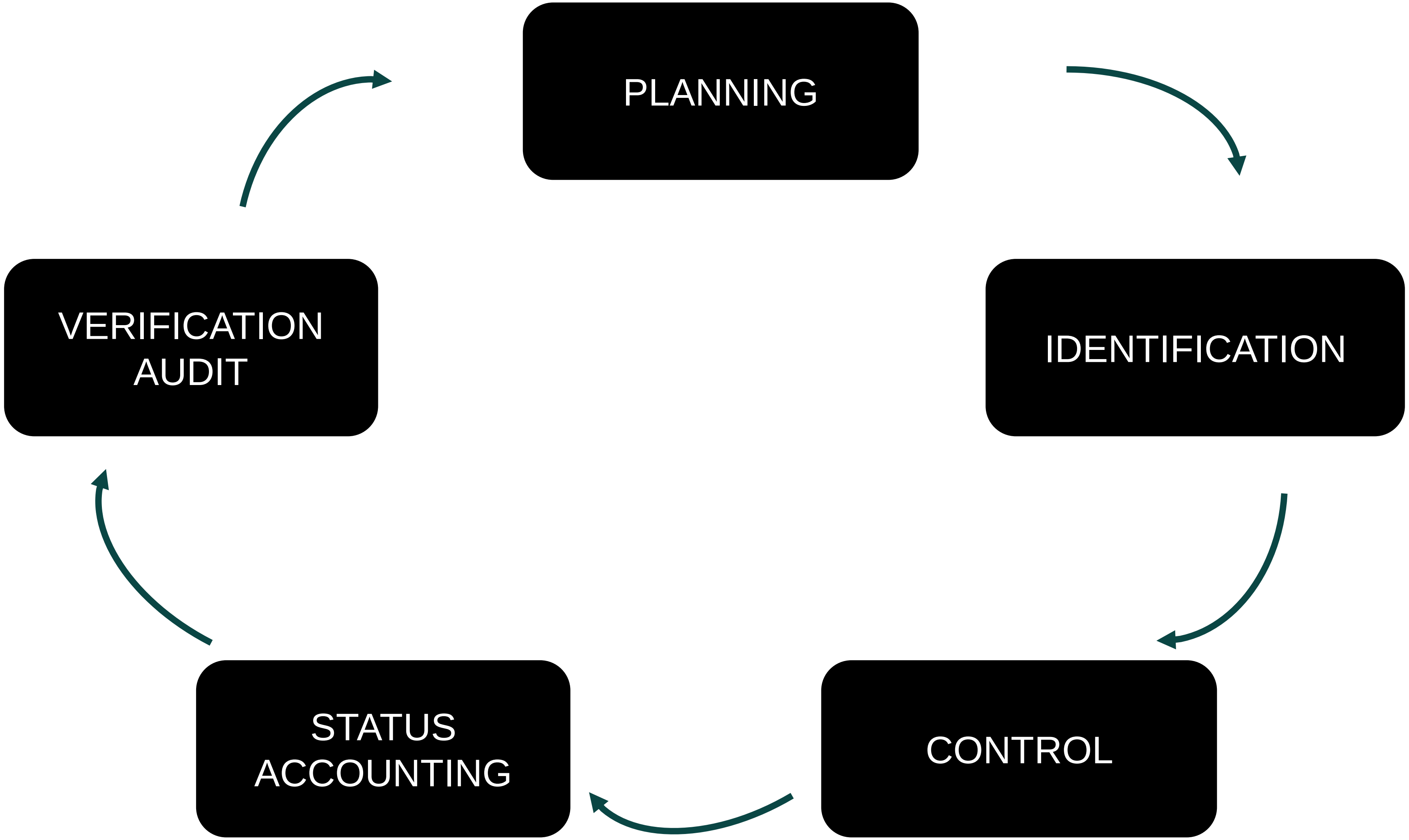
Must have some flexibility
What can be sacrificed if time and cost are threatened?

MUST
O
SHOULD
COULD
O
WON'T



16b) Understand how to manage scope through configuration management processes (such as planning, identification, control, status accounting and verification audit).

Configuration management process



Configuration management process

Scope management is closely linked to configuration management by planning, protecting and controlling changes to requirements. The activities of configuration management contribute to scope management as follows:

Planning

- Defines process and procedures, along with setting clear roles and responsibilities to communicate requirements.

Identification

- Divides the project into configurable items, each with a unique reference so that the project manager and team can identify the deliverables of the project.

Control

- Seeks to ensure that changes to these items are controlled and interrelationships between the items are understood to fully understand the impact of the change.

Status accounting

- Ensures traceability of configuration items throughout their development. This supports record keeping and document control for the project, thus maintaining a robust audit trail of configuration records over the course of the project.

Verification audit

- This ensures that the deliverable conforms to its requirements and configuration information, and that completed deliverables match those documented. Audits should be undertaken throughout the life cycle to maintain full end-to-end configuration management of the project.